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Strategic positioning of the order penetration point

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Abstract

The order penetration point (OPP) defines the stage in the manufacturing value chain, where a particular product is linked to a specific customer order. Different manufacturing environments such as make-to-stock (MTS), assemble-to-order (ATO), make-to-order (MTO) and engineer-to-order all relate to different positions of the OPP. These may be considered as product delivery strategies, having different implications for manufacturing objectives such as customer service, manufacturing efficiency and inventory investment. Furthermore, the OPP may differ between products and over time for a particular manufacturing firm. In this paper, the positioning of the OPP is treated from a strategic perspective. Market, product, and production factors are identified that affect the OPP positioning and the shifting of the OPP upstream or downstream in the manufacturing value chain. The major factors are demand volume and volatility, and the relationship between delivery and production lead times. These factors are included in a model that allows the manufacturing firm to choose the right product delivery strategy. Different manufacturing strategies must be developed for pre-OPP operations (i.e. upstream; forecast-driven) vs. post-OPP operations (i.e. downstream, customer-order-driven), since these two stages are fundamentally different. As a consequence, a manufacturing firm that has an ATO product delivery strategy must differentiate between MTS operations (upstream the OPP) and MTO operations (downstream the OPP). For example, the competitive priorities differ: price for pre-OPP operations but delivery speed and flexibility for post-OPP operations. Therefore, decision categories, such as production planning and control, and performance measurement must be designed accordingly. Guidelines are provided for this strategic choice.
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1. Introduction

The positioning of the order penetration point (OPP) is successively becoming a topic of strategic interest. With global markets, increasing global competition and shorter product life cycles, the choices and shifts between make-to-order (MTO) and make-to-stock (MTS) policies must be made faster and at a strategic level. The motivation for this paper is to investigate factors that affect the positioning and shifting of the OPP. Of special interest is the rationale of forward shifting, something that the author has experienced in a few manufacturing firms. Backward shifting is generally assumed to be desirable to reduce the number of activities that are planned using uncertain information, i.e. forecast-driven, potentially reducing or eliminating inventories.

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Berry and Hill (1992) refer to the OPP choice as that of choosing a master planning approach, distinguishing between MTS, assemble-to-order (ATO), and MTO. This approach is further developed in Vollmann et al. (1997) and Hill (2000). The discussion in the literature on the strategic importance of the positioning of the OPP is very sparse. Sharman (1984) introduced the term OPP in a logistics context. He defines the OPP as the point where product specifications typically get frozen, and as the last point at which inventory is held. He argues that the OPP depends on a balance between competitive pressure and product cost and complexity. Olhager and Östlund (1990) discuss the use of push and pull systems relative the position of the OPP, arguing that pull-type systems are applicable upstream of the OPP and push-type systems are necessary for downstream operations. Their analysis also included the bottleneck position and product structure as factors influencing the integration of push and pull systems.

In this paper, we try to add factors and perspectives to derive a more holistic view of the strategic relevance of choosing the right product delivery strategy, i.e. choosing the right OPP position. First, we define the OPP and discuss its characteristics. The factors that affect and to some extent constrain the position of the OPP are categorised and a conceptual impact model is developed. The OPP is then related to a manufacturing strategy, implying that a distinction is necessary between pre-OPP and post-OPP operations, since these have fundamentally different characteristics. Then, the reasons and the means for redesigning or rebalancing the manufacturing value chain, by shifting the OPP backwards or forwards, are explored. Finally, a model is presented for choosing the right product delivery strategy.

2. Characteristics of the OPP

The OPP is traditionally defined as the point in the manufacturing value chain for a product, where the product is linked to a specific customer order. Sometimes the OPP is called the customer order decoupling point (CODP) to highlight the involvement of a customer order. Different manufacturing situations such as MTS, ATO, MTO and engineer-to-order (ETO) all relate to different positions of the OPP. The different manufacturing situations are related to the ability of the manufacturing operations to accommodate customising or a wide product range (see Fig. 1). Thereby, the OPP divides the manufacturing stages that are forecast-driven (upstream of the OPP) from those that are customer-order-driven (the OPP and downstream).

3. Factors affecting the positioning of the OPP

The positioning of the OPP is affected by many factors; see e.g. Sharman (1984), Olhager and Östlund (1990), Giesberts and Van der Tang (1992), Hoekstra and Romme (1992), Olhager (1994), Van der Vlist et al. (1997), Fisher (1997), Olhager and Wikner (1998, 2000), Pagh and Cooper (1998), Lehtonen (1999a, b), D'Alessandro and Baveja (2000), Mason-Jones et al. (2000), and Van Donk (2001). The majority of the literature deals with manufacturing operations, whereas Sharman (1984), Hoekstra and Romme (1992), Fisher (1997), Pagh and Cooper (1998), and Mason-Jones et al. (2000) primarily deal with supply chains. Case examples are found in Olhager and Östlund (1990), Van der Vlist et al. (1997), Lehtonen (1999a, b), D'Alessandro and Baveja

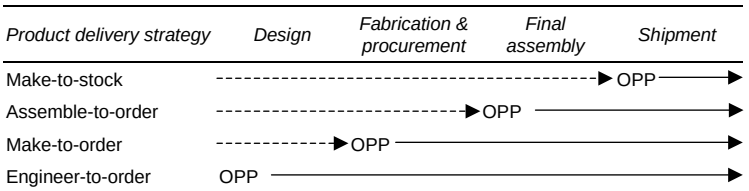


Fig. 1. Different product delivery strategies relate to different order penetration points. The dotted lines depict the production activities that are forecast-driven, whereas the straight lines depict customer-order-driven activities.

(2000), and Van Donk (2001), concerning printed packaging boxes, trucks, pulp and paper, chemicals, and food processing, respectively. Some authors do not explicitly use the term OPP (or CODP) but discuss MTS, ATO and MTO in a way that is fruitful to the understanding of the characteristics and impact of the OPP position. We divide the most important factors into three categories, related to (i) market, (ii) product, and (iii) production characteristics.

3.1. Market-related factors

Delivery lead-time requirements set by the market restricts how far backwards the OPP can be positioned. Furthermore, this is a benchmark for production lead-time improvements in order to make delivery speed an order winner.

Product demand volatility indicates to what extent it is possible or reasonable to make products to order or to stock. Low volatility means that the item can be forecast-driven. However, high volatility makes forecasting difficult, wherefore such items typically need to be produced to order.

Product volume is related to demand volatility in that the relative volatility is lower for high-volume items. This can be measured by the coefficient of variation, i.e. the ratio between the standard deviation of demand and average demand. Demand volumes increase during the initial stages of the product life cycle, wherefore the order winners may well change from delivery speed, design, and flexibility to price. At some point in time, the product may be regarded as a commodity. At the end of the product life cycle volumes go down, potentially leading to a reversal to the initial order winner criteria.

Product range and product customisation requirements are similar in nature with respect to the OPP. A broad product range and a wide set of customisation required by the customer would be impossible to provide on an MTS basis. The investment in final goods inventory would be too immense. A narrow range and predetermined customer choices make it possible to move towards ATO or even MTS.

Customer order size and frequency are indicators of volume and the repetitive nature of demand. Large customer order sizes are typically associated with high demand volumes. It furthermore indicates that there typically is an intermediary part between the producer and ultimate consumer such as a retailer or wholesaler buying in batches. High frequency leads to repetitive demand making forecasting easier. These two factors can be interrelated such that a customer makes a long-term delivery contract (large total order size) with frequent small-lot call-offs (high frequency).

For products with highly *seasonal demand*, it may be uneconomical for the manufacturing firm to respond to all demand when it occurs. Consequently, the firm may choose to manufacture some products to stock in periods with low demand in anticipation of peak demand. Thereby, production is levelled and plant utilisation increases. Thus, a product may shift between MTS and MTO or ATO depending upon the season. In some cases, it may be sufficient for the firm to manufacture fabricated parts and sub-assemblies during periods with low demand and then assemble the end products during demand peaks. Preferably, these items have high work content relative to material usage, whereas the items that are manufactured during high demand periods have a high material content and limited work content, thereby improving the flexibility to respond to actual customer orders with short delivery lead times.

3.2. Product-related factors

Modular product design is typically related to ATO product delivery strategies. Such initiatives are often a response by the producer to create a variety of choices for the customer, a relatively short delivery lead time, and manufacturing efficiency for upstream operations.

The *customisation opportunities* that the producer provides in the product design in anticipation of the customisation requirements is associated with anticipated product delivery strategies. If the customisation offered is wide and enters the product at early production stages, an MTO policy is necessary, whereas if customisation enters

at a very late production stage ATO may be more appropriate.

The number of items at various levels of the product structure, which may be represented by various letters, constitutes the *material profile*. The top, middle and bottom depict the number of end products, sub-assemblies (or fabricated parts) and components (or raw materials), respectively. Umble and Srikanth (1990) describe V, A and T profiles and the planning problems associated with those profiles, whereas Olhager and Wikner (1998) focus on V, A, and X profiles for material and capacity-dominated master scheduling. V is the typical profile for a process firm with a divergent material flow from raw materials to finished goods. A and T profiles represent assembly situations. An A firm has a successive assembly of parts at many product structure levels, whereas a T firm can assemble a large variety of end products from a narrow set of pre-defined modules, cf. the concept of postponement. An X profile is the typical result of a modular product design, where upstream operations are made to stock and downstream operations create the customer-specific product, based upon the choice of modules at the OPP, which is positioned at the material profile waist. Thus, an X profile is built up by a V profile on top of an A profile.

The breadth and depth of the *product structure* indicates the product complexity. A deep product structure may well correspond to long cumulative production lead times. Then the various paths of the product structure need to be analysed in terms of lead times to determine where in-process inventories need to be kept relative the delivery lead-time requirements.

3.3. Production-related factors

The *production lead time* is a major factor to consider with respect to the delivery lead-time requirements set by the market. This relationship poses a major constraint on the OPP position. Lead-time reductions can then relieve such a constraint, and lead to a wider range of OPP positioning opportunities.

The number of *planning points* in a manufacturing process restricts the number of potential OPP positions. A planning point is a manufacturing resource or a set of manufacturing resources such as a work centre or a work cell that can be regarded as one entity from a production and capacity planning point of view. In a job shop where individual resources are planned the variety for positioning the OPP is large. A dedicated line or continuous process can be treated as a single production unit and therefore offers only two possibilities: before or after the process.

The *flexibility* of the production process, e.g. through short set-up times, is a prerequisite for producing to order. Thereby, a wider range of products and customisation can be accommodated in the production system.

The position of the *bottleneck* of the production process relative that of the OPP is interesting. From a resource optimisation point of view, it is advantageous to have the bottleneck upstream the OPP, so the bottleneck does not have to deal with volatile demand and a variety of different products. With respect to the just-in-time principle of elimination of waste, it would be best to have the bottleneck downstream the OPP so that the bottleneck only needs to work on products for which the firm has customer orders. A bottleneck can be a candidate for the OPP, especially if it is an expensive resource performing significant activities in the production process of the product; cf. the case in Olhager and Östlund (1990).

Resources with *sequence-dependent set-up times* are best positioned upstream the OPP. Such resources can easily turn into bottlenecks without proper sequencing, which is a likely course of action for downstream operations.

3.4. A conceptual factor impact model

There are many factors that potentially can affect the position of the OPP, as described above. They are interrelated to some extent, as illustrated in Fig. 2. Market characteristics affect product characteristics. The product range and customisation opportunities interact with market expectations and result in a delivery lead time that customers require with respect to the product offering. These factors are input to the production function. The product structure can be interpreted

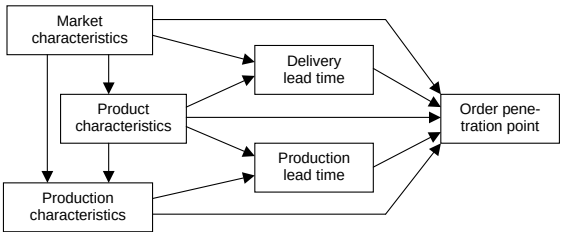


Fig. 2. Conceptual impact model for factors affecting the positioning of the OPP.

in terms of lead time with respect to the operations that need to be performed at each level. The relationship between production and delivery lead times is a major determinant of the OPP position.

4. The role of the OPP in a manufacturing strategy

The manufacturing strategy is strongly related to the marketing and product strategy, and can be viewed in two dimensions (Leong et al., 1990): competitive priorities and decision categories. Manufacturing-based competitive priorities include price, quality, delivery speed and reliability, and flexibility in terms of volume and product mix, see e.g. Hayes and Wheelwright (1984), Miltenburg (1995), and Hill (2000). Hill introduced the concept of order winners and qualifiers in order to differentiate the importance among competitive priorities. An order winner makes the product win orders in the market place, whereas qualifiers are criteria that must be provided by the firm in order to enter or stay in the market (Hill, 2000). The other dimension, decision categories, constitutes the framework for formulating a manufacturing strategy, see e.g. Hayes and Wheelwright (1984), Miltenburg (1995), and Hill (2000). A manufacturing strategy consists of policies for areas and issues related to the decision categories: process, capacity, facilities, vertical integration, quality, organisation, production planning and control, and performance measurement.

Fig. 3 highlights the most important issues concerning the market, product and production attributes, illustrating that there is a fundamental difference between pre-OPP and post-OPP operations. Mason-Jones et al. (2000) distinguish between lean and agile supply chains, where a lean supply chain would be applied upstream the OPP, whereas an agile supply chain would be more suitable for downstream operations. They acknowledge the OPP as the divider between these two supply chain types. Fisher (1997) makes a distinction between physically efficient and market-responsive supply chains. The lean and agile supply chains in Mason-Jones et al. (2000) resemble the physically efficient and the market-responsive supply chains in Fisher (1997), respectively, but Mason-Jones et al. (2000) put them in context relative to a decoupling point. However, the concept of a manufacturing strategy is not used in their supply chain contexts.

The main competitive priority that is directly related to the position of the OPP is delivery speed. If delivery speed is an order winner, the OPP should be positioned closer (in terms of time) to the final goods inventory than that of the competitors. Customers expect some products to be available off the shelf at any time, forcing the manufacturer to an MTS policy. The customer order process enters the manufacturing value chain at the OPP, wherefore post-OPP operations must be competitive with respect to delivery speed. Pre-OPP operations on the other hand focus on maintaining an optimal mix and optimal inventory levels at the OPP. Since pre-OPP operations are forecast-driven and need not focus on delivery speed, resource capacity can be reduced and optimised. Thus, focus moves to price competition via cost efficiency with respect to capital tied up in capacity and inventories. Should these resources be customer-order-driven instead, the possibility to optimise the process is in general drastically reduced, since it has to respond to the variability of customer demand, requiring some excess capacity.

With respect to the decision categories in a manufacturing strategy, the differences are distinct between pre-OPP and post-OPP operations. For example, a flow-oriented production process is more applicable for upstream operations since the number of products is limited (i.e. a product layout and a product facility focus), whereas the downstream operations need the flexibility inherent in a job shop (i.e. a process layout and a process facility focus). Since the replenishment

Attributes	*Pre-OPP operations*	*Post-OPP operations*
Markets and products		
Product type	Standard, commodities	Special
Product range	Predetermined, narrow	Wide
Demand	High volume, predictable	Low volume, volatile
Order winners	Price	Design, flexibility, delivery speed
Market qualifiers	Design, quality, on-time delivery	Price, quality, on-time delivery
Production (decision categories)		
Process	Line, high-volume batch	Job shop, low-volume batch
Capacity	Lag/track	Lead/track
Facilities	Product focus	Process focus
Vertical integration	Supplier relationships, OPP buffer/post-OPP operations	Customer relationships, OPP buffer/pre-OPP operations
Quality	Process quality focus	Product quality focus
Organisation	Centralised	Decentralised
Production planning and control	Level S&OP strategy Order promising based on stock availability Rate-based material planning Pull-type execution	Chase S&OP strategy Order promising based on lead time agreement, and material and capacity availability Time-phased material planning Push-type execution
Performance measurement	Cost, productivity	Flexibility, delivery lead times

Fig. 3. Comparison of manufacturing strategy attributes for pre-OPP vs. post-OPP operations.

process of the inventory at the OPP is controlled by the manufacturing firm, a lag capacity strategy can be managed. For downstream operations, a lead-type capacity strategy is necessary to avoid long and variable customer order lead times; if capacity is tight for the customer-order-driven activities, demand volatility will transfer to lead-time volatility. Since the OPP provides a decoupling point between upstream and downstream operations, the perspectives are extended to the outside vertically integrated contacts, i.e. to suppliers upstream and customers downstream, aiming at providing efficient links to the closest internal activities through collaborative efforts. Quality planning, control and improvement will focus on maintaining efficient processes for upstream operations, whereas the product delivered to the customer is the main planning object for post-OPP operations. The organisational type can be centralised upstream, whereas a decentralised approach is more appropriate for downstream activities focussing on delivering a customised product on time. The design of the production planning and control system follows the principles in Berry and Hill (1992), extended by Olhager et al. (2001) for the sales and operations planning level. Performance measurement needs to focus on the order winners and market qualifiers for each respective part around the OPP. This means that cost and productivity are important performance measures for upstream operations since price is the dominant order winner, whereas downstream operations need to measure the relevant means of flexibility (e.g. product mix and volume flexibility; cf. Olhager (1993) and Olhager and West (2002)) and delivery lead times for competing on design, flexibility, and delivery speed. Taking these differences into consideration it is apparent that there is a need for manufacturing firms to differentiate between pre-OPP and post-OPP operations.

5. Forward and backward shifting

Any shift in the OPP needs to be strategically motivated, i.e. it should strengthen a competitive priority. A major possibility for strengthening the competitive edge is through a lead-time reduction, preferably for downstream activities. Two options are then available. The firm can either maintain the OPP at the current position and reduce the delivery lead times or maintain the delivery lead time and shift the OPP backwards, to reduce or eliminate some in-process inventories.

If the market changes in terms of how products win orders in the market place, then the OPP should be shifted accordingly. This is especially true for delivery speed. If delivery speed can be made into an order winner in a market where it used to be a qualifier, then the market requirements as to deliveries have changed, and a firm can take advantage based on this knowledge, e.g. by shifting the OPP forwards to reduce the delivery lead time or by simultaneously reducing the production lead time accordingly.

In the next two sections, we explore the strategic issues, the reasons to shift and the potential negative effects of shifting the OPP forwards or backwards, respectively. These sections exemplify the trade-offs when considering a shift in the OPP in practice.

5.1. Forward shifting

There are two major driving forces for shifting the OPP forward: (i) reduce the delivery lead time to customers, and (ii) increase the manufacturing efficiency, e.g. by optimising the bottleneck operation. For example, in the case of sequence-dependent set-up times on a bottleneck, it would be advantageous from an optimisation point of view to be able to run this resource based on a forecast, rather than having to react to exact customer order items and quantities. These two driving forces address delivery speed and reliability, and price, respectively.

The major means for forward shifting is to increase the level of pre-fabrication, e.g. by introducing a modular design for products. A good modularisation will typically lead to an ATO policy. The reduction of lead times for downstream activities will reduce the delivery lead time (which is a reason for forward shifting), but will not change the OPP with respect to the material flow in the product structure per se. On the contrary, lead-time reduction will typically be used for shifting the OPP backwards. The reasons for shifting the OPP forwards along with the associated negative effects and the strategic issues addressed are summarised in Fig. 4.

5.2. Backward shifting

The main driving forces for shifting the OPP backwards is to increase the knowledge of the contents of customer orders at the time of production, i.e. allowing a higher degree of customisation and to reduce the amount of work-in-process inventory. By performing more activities based on actual customer order quantities, the organisation will focus on quality, since the activities during the delivery lead time must yield

Competitive advantage addressed	*Reasons for forward shifting*	*Negative effects*
Delivery speed Delivery reliability Price	Reduce the customer lead time Process optimisation (improved manufacturing efficiency)	Rely more on forecasts (risk of obsolescence) Reduce product customisation (to maintain WIP and inventories levels) Increase work-in-process (due to more items being forecast-driven)

Fig. 4. Strategic issues, reasons and negative effects of shifting the OPP forwards.

Competitive advantage addressed	*Reasons for backward shifting*	*Negative effects*
Product range Product mix flexibility Quality	Increasing the degree of product customisation Reduce the reliance on forecasts	Longer delivery lead times and reduced delivery reliability (if production lead times are not reduced)
	Reduce or eliminate WIP buffers Reduce the risk of obsolescence of inventories	Reduced manufacturing efficiency (due to reduced possibilities to process optimisation)

Fig. 5. Strategic issues, reasons and negative effects of shifting the OPP backwards.

sufficient quality for making the delivery complete. Also, the risk for obsolete materials in in-process inventories will reduce.

The major means for backward shifting is lead-time reduction of downstream activities in order to maintain the delivery lead time while performing more production activities within the delivery lead time, and thus being able to cope with a higher degree of customer specifications. Reasons for shifting the OPP backwards along with the associated negative effects and the strategic issues addressed are summarised in Fig. 5.

6. Perspectives on the positioning of the OPP

The two major factors that affect the strategic positioning of the OPP are the production to delivery lead time ratio (P/D ratio) and the relative demand volatility (RDV). The RDV is defined as the coefficient of variation, i.e. the standard deviation of demand relative the average demand. If we juxtapose these two factors in a diagram, we can identify four basic situations. Both these factors can range from low to high, leading to four combinations with different properties pertaining to the choice of a product delivery strategy.

An interesting cut-off point for the P/D ratio occurs at the value of one. This value is related to whether production can await a customer order before production activities must begin or not. If not, at least some activities must be based on forecasts. If the P/D ratio is less than one, an MTO product delivery strategy is possible. However, in order to capitalise on economies of scale for common items, these may be produced to stock. This is expressed through the RDV, such that a low RDV indicates that some parts can be produced to stock, potentially leading to ATO situation. For very low RDVs MTS strategy might even be pursued. This means that even though the P/D ratio would allow for MTO, the firm chooses an MTS policy with the purpose to increase productivity. If, on the other hand, the RDV is high, an MTO policy is the natural choice. If the P/D ratio is larger than one, part of the internal supply chain must be forecast-driven. Thus, an MTO policy is impossible, forcing the product delivery policy to be either ATO or MTS instead. If the RDV is high it is not reasonable to use an MTS policy, since this would lead to excessive inventories. Therefore, products will be ATO. It is then of the utmost importance to identify the right OPP position. A balance must be made between on the one hand minimising the number of forecast items and, on the other, maximising the opportunities to take advantage of economies of scale. One restriction is that the operations that are made-to-order must be contained within the delivery lead time. Individual items that are outside the delivery lead time and have high individual RDVs pose some planning problems. At the same time, individual items within the delivery lead time that have low individual RDVs provide opportunities for increasing productivity.

A complementary explanation to Fig. 6 refers to the manufacturing response to market and demand characteristics. Firstly, if the market

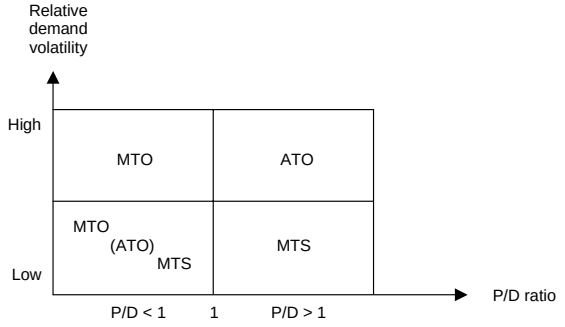


Fig. 6. A model for choosing the right product delivery strategy. P/D ratio is the production lead time to delivery lead time ratio, and the RDV is the coefficient of variation, i.e. the standard deviation of demand relative average demand.

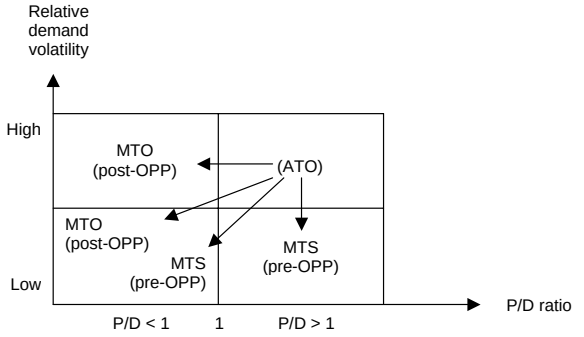


Fig. 7. Decoupling ATO into MTS for pre-OPP operations and MTO for post-OPP operations. MTO is only possible for P/D ratios less than one. For low RDV, employing an MTS policy can capture economies of scale.

requires a short delivery lead time, pressure is put on manufacturing to reduce lead times to qualify for a wider choice of product delivery strategies. If the manufacturing lead time is long, MTO will never be a choice. A manufacturing firm that proactively cuts lead times may create opportunities where delivery speed can be made an order winner, at least in some markets for some products. Such action resembles the ultimate strategic role of manufacturing, i.e. to be externally supportive, according to Hayes and Wheelwright (1984). Secondly, if demand variability is too high for using an MTS policy, inventory must be kept at some point along the internal manufacturing value chain. This point would be where the coefficient of variation, i.e. the RDV, is reasonably small. Typically, a lower RDV coincides with higher demand volume, i.e. the higher the product volume, the lower the relative demand fluctuation. This has been shown empirically in D'Alessandro and Baveja (2000). Olhager (2002) relates this fact to the product structure, showing that demand volumes are non-decreasing down through the product structure. This means that the opportunities for MTS policies increases for upstream operations. What is judged to be a "reasonably" small RDV in this context may differ between firms and is related to the inventory investment associated with the MTS option.

Next, we will pay special attention to the upper right-hand corner, characterised by a high P/D ratio and high RDV. This quadrant is the least desirable. Here ATO is employed to deal in a reasonably good way with volatile demand and delivery lead-time requirements. By viewing the ATO option as a combination of MTS upstream and MTO downstream the OPP, it is possible to reposition ATO in Fig. 6. Fig. 7 shows the alternatives available for applying pure MTS and MTO policies for pre-OPP and post-OPP operations, respectively.

By splitting the ATO option into two parts it may well occur that each part will have a lead time that is less than the delivery lead time. By definition, MTO must correspond to a P/D ratio that is less than one. MTS becomes a viable option for any case with low RDV. However, smaller P/D ratios in combination with larger RDV make MTO successively more appropriate. Still, it must be remembered that when decoupling an ATO situation around the OPP, the result must be MTS upstream and MTO downstream the OPP, i.e. both policies must be employed. We are not considering a complete shift to MTO. Therefore, it is stressed in Fig. 7 that MTO is only applicable to post-OPP operations.

7. Discussion

In this paper, we have presented several factors that affect the position of the OPP, and discussed

the strategic rationale of shifting the OPP forwards or backwards. Extreme backward shifting will place the firm in an ETO situation, where design, delivery speed and flexibility are typical order winners. On the other extreme, forward shifting as far as possible leads to an MTS situation where price is the dominating order winner. Such shifts are related to the competitive advantages in terms of order-winning criteria and to the extent these need to be strengthened or must be realigned due to a change in the market. There is a trade-off between (i) maximum manufacturing efficiency that dominates the pre-OPP operations and (ii) minimum inventory investment that dominates the post-OPP operations, while at the same time maintaining a high and consistent level of customer service. In this context, the positioning of the OPP is the very core of this trade-off. The OPP divides the manufacturing value chain into two separate parts that must be differentiated with respect to manufacturing strategy. This implies that two manufacturing strategies must be developed: one for pre-OPP operations and one for post-OPP operations. We presented some basic strategic issues associated with pre-OPP vs. post-OPP operations. We also presented a model that can assist in identifying the proper product delivery strategy, with respect to demand volume and volatility, and the relationship between the production and delivery lead times. In conclusion, we hope that this paper can contribute to the understanding of the OPP, its characteristics and strategic importance.

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